

The atomless band projection is not a positive self-commutator

Candidate full negative answer for arXiv:2501.03733

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Status: candidate full negative answer, likely valid. On the Hilbert lattice $\ell^2 \oplus L^2[0, 1]$, the projection $0 \oplus I$ cannot equal $A^*A - AA^*$ for a positive operator A . In fact, no *regular* operator A has that self-commutator.

1 Source question

The source is R. Drnovšek and M. Kandić [1]. Their article, *Positive self-commutators of positive operators* (arXiv:2501.03733), appeared in *Positivity* 29 (2025), Article 43. Its concluding Question 5.3 asks whether, on the Hilbert lattice

$$H = \ell^2 \oplus L^2[0, 1],$$

the positive operator

$$C = \begin{pmatrix} 0 & 0 \\ 0 & I \end{pmatrix} \tag{1}$$

is a self-commutator $A^*A - AA^*$ of a positive operator A .

$$\setminus \begin{pmatrix} 0 & 0 & 0 & 0 \end{pmatrix}$$

is a self-commutator of the positive operator $A \oplus B$. □

We conclude this paper with the following question.

Question 5.3. Is the operator C on the Hilbert lattice $\ell^2 \oplus L^2[0, 1]$ represented with a 2×2 block operator matrix

$$\begin{pmatrix} 0 & 0 \\ 0 & I \end{pmatrix}$$

a self-commutator of a positive operator?

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Figure 1: Question 5.3 on page 17 of the official source PDF.

2 Regular kernel operators

We isolate the standard Banach-lattice input. For a Banach lattice E , write $\mathcal{L}^r(E)$ for the unital Banach algebra of regular operators on E , equipped with the regular norm. On $L = L^2[0, 1]$, let

$$J = \mathcal{I}^r(L)$$

be the band of regular kernel operators, equivalently the band generated by the regular finite-rank operators.

Lemma 1. *The set J is a proper closed two-sided ideal of $\mathcal{L}^r(L)$. Moreover, every regular operator on L that regularly factors through ℓ^2 belongs to J .*

Proof. The band J is norm closed. Because L has order-continuous norm, the standard composition properties of regular kernel operators show that it is a two-sided algebra ideal in $\mathcal{L}^r(L)$.

For the factorization assertion, use Blanco's Theorem 2(ii) [2]: if a regular operator between Banach lattices regularly factors through an order-continuous purely atomic Banach lattice, then it belongs to the band generated by the regular finite-rank operators. The lattice ℓ^2 is order continuous and purely atomic, so the theorem applies.

For completeness, J is proper because I_L is not a kernel operator. Indeed, if a measurable kernel $k(t, x)$ represented I_L , then, after testing the constant function, $k(\cdot, x)$ would be integrable for almost every x . Testing the indicators of a countable algebra generating the Lebesgue measurable sets would give

$$\int_A k(t, x) dt = \mathbf{1}_A(x)$$

for all sets A in that algebra and almost every fixed x . By uniqueness of finite measures this says that the absolutely continuous measure $k(t, x) dt$ is the point mass δ_x , impossible on the atomless Lebesgue space. Hence $I_L \notin J$. $\square$

3 The negative answer

Theorem 1. *There is no regular operator A on $\ell^2 \oplus L^2[0, 1]$ such that*

$$A^*A - AA^* = 0 \oplus I.$$

Consequently, the answer to Question 5.3 is negative.

Proof. Suppose such a regular operator exists and write it in blocks as

$$A = \begin{pmatrix} P & Q \\ R & S \end{pmatrix}, \quad Q : L \longrightarrow \ell^2, \quad R : \ell^2 \longrightarrow L.$$

The coordinate inclusions and projections are positive, so all four blocks are regular. Adjoints of regular operators are regular as well. Comparing the lower-right corners of $A^*A - AA^*$ and (1) gives

$$I_L = Q^*Q - RR^* + S^*S - SS^*. \quad (2)$$

Both cross terms in (2) regularly factor through ℓ^2 :

$$L \xrightarrow{Q} \ell^2 \xrightarrow{Q^*} L, \quad L \xrightarrow{R^*} \ell^2 \xrightarrow{R} L.$$

Lemma 1 therefore puts Q^*Q and RR^* in J . Since $I_L \notin J$, the quotient

$$\mathcal{A} = \mathcal{L}^r(L)/J$$

is a nonzero unital Banach algebra. If $s = S + J$ and $t = S^* + J$, equation (2) descends to

$$1_{\mathcal{A}} = ts - st = [t, s]. \quad (3)$$

No unit of a Banach algebra is a commutator. Here is the short Wintner argument. From $[t, s] = 1$ one obtains inductively

$$[t, s^n] = ns^{n-1} \quad (n \geq 1).$$

Thus

$$n \|s^{n-1}\| \leq 2\|t\| \|s^n\|,$$

and iteration yields

$$\|s^n\| \geq \frac{n!}{(2\|t\|)^n}.$$

This contradicts $\|s^n\| \leq \|s\|^n$ as $n \rightarrow \infty$. Hence (3), and therefore the assumed A , cannot exist.

Every positive operator is regular, so the stronger statement implies the negative answer requested by the source. $\square$

4 Verification, scope, and novelty

The proof uses only the lower-right block equation; no unproved assertion about the other three blocks is needed. The signs in (2) follow from

$$(A^*A)_{22} = Q^*Q + S^*S, \quad (AA^*)_{22} = RR^* + SS^*.$$

The imported factorization theorem applies to arbitrary regular factors, so positivity is not needed for the strengthening. A detailed adversarial audit appears in the companion `verification.md`.

The local run indexes were searched using the arXiv identifier and exact title, as well as terms for the mixed atomic/atomless projection, positive self-commutators, and regular kernel factorization. Bounded web searches on 11 August 2026 used the exact question and the kernel-ideal quotient route. They found the source and the background factorization theorem but no later answer to Question 5.3 and no matching quotient proof. Novelty confidence is moderate because citation and web coverage is bounded.

References

- [1] R. Drnovšek and M. Kandić, *Positive self-commutators of positive operators*, Positivity 29 (2025), Article 43; arXiv:2501.03733. <https://arxiv.org/abs/2501.03733>.
- [2] A. Blanco, *Ideal structure and factorization properties of the regular kernel operators*, Positivity 24 (2020), 1211–1229. <https://doi.org/10.1007/s11117-019-00728-7>.